Lecture 12: Mesh Current Analysis

EE201 – Circuit Theory I

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current, mesh current 1 voltage supermesh

Review: node voltage analysis

The procedure

- Choose the ground; label the node voltages $\mathbf{e}$.
- KVL holds automatically: $\mathbf{v} = A^T \mathbf{e}$.
- Write KCL at every node except the ground.
- Matrix form: $Y_n \mathbf{e} = \mathbf{i}_n$.

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- From the ground to node k: e_k is known.
- Between two nodes: a supernode.
- A dependent source: one more constraint.

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Can we do the opposite?

Motivation

Node voltages

- Unknowns: the $n-1$ node voltages $\mathbf{e}$.
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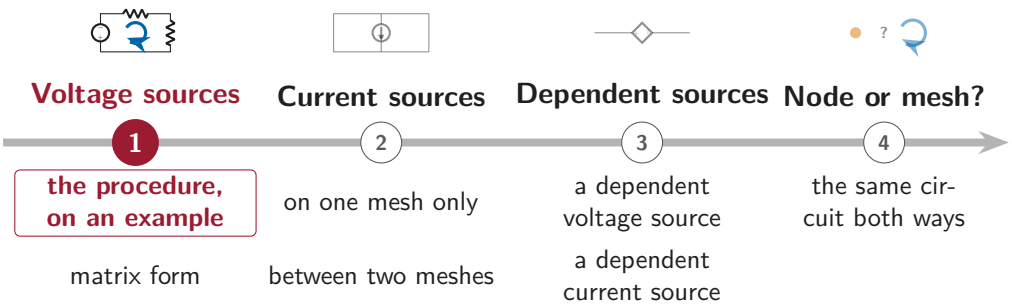
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Mesh current analysis follows the same plan, with KCL and KVL swapped.

Roadmap

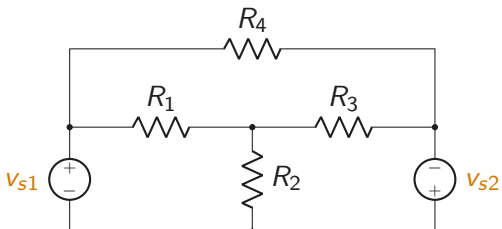


Each station adds a new kind of source; the last one compares the two methods.

Example 12.1: mesh current analysis step by step

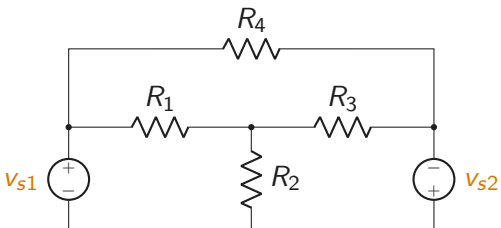
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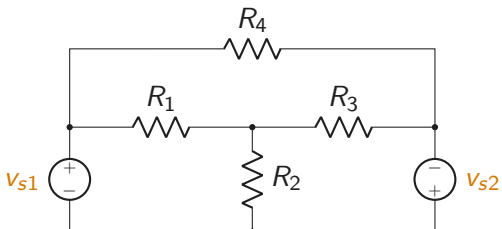


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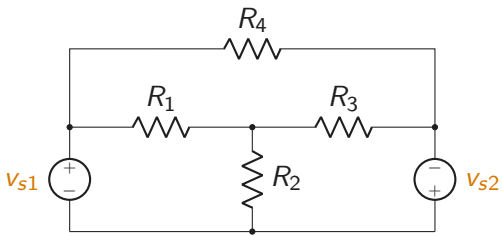
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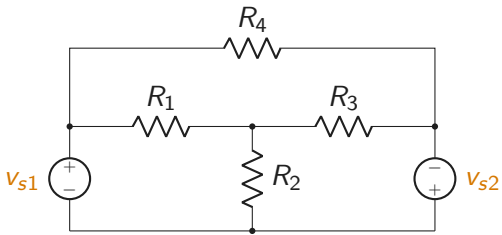
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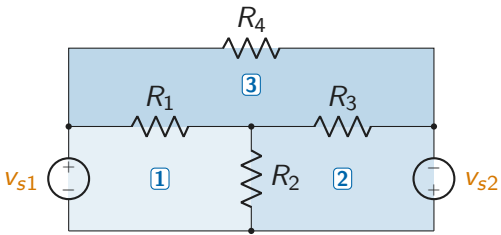
We find them in four steps, on this circuit.

Step 1: find the inner meshes



The circuit is planar: it is drawn with no crossing branches.

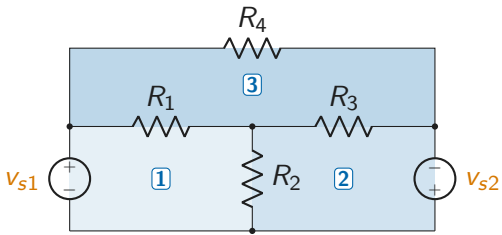
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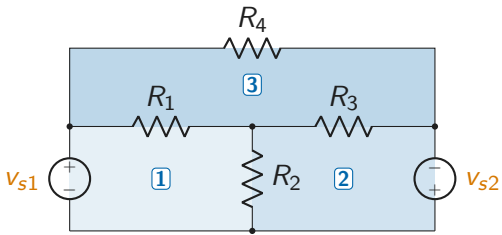


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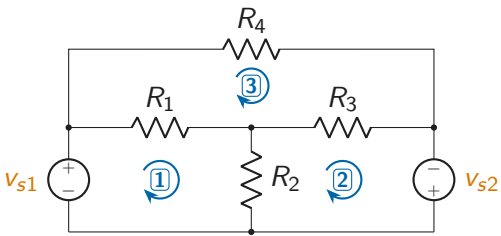
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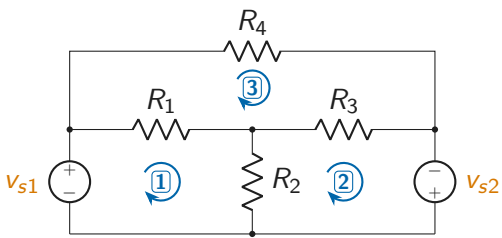
Mesh current analysis has one unknown per inner mesh.

Step 2: assign the mesh currents



Each inner mesh gets a mesh current, clockwise: i_{M1}, i_{M2}, i_{M3}.

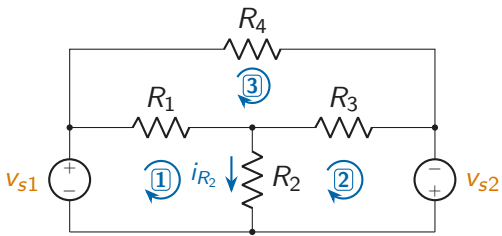
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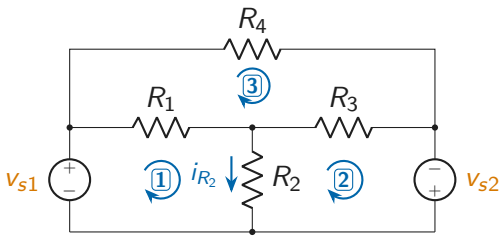
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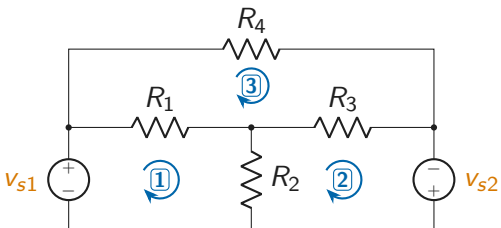
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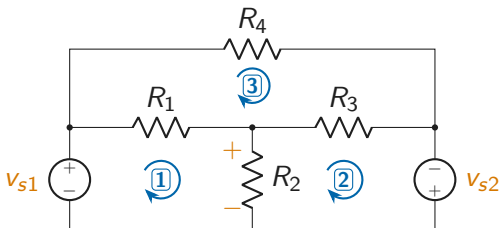
A mesh current enters and leaves every node it passes, so KCL holds automatically.

Step 3: write the resistor voltages with mesh currents



Ohm's law gives each resistor voltage from its mesh currents.

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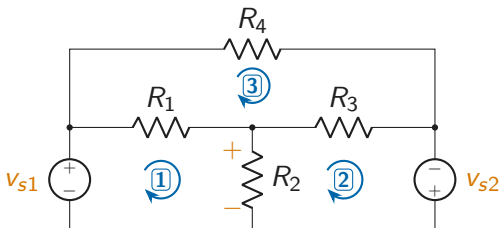


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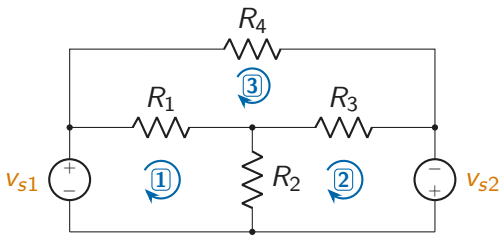
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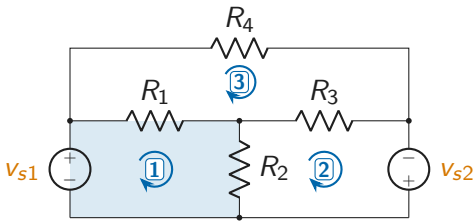
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The rule: R times (own mesh current minus the other one).

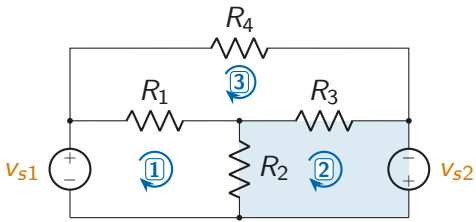
Step 4: KVL around every inner mesh



Mesh 1:

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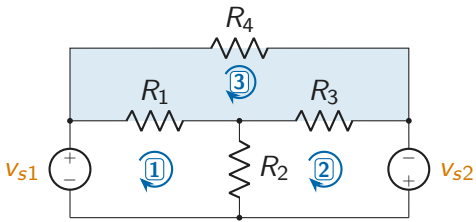
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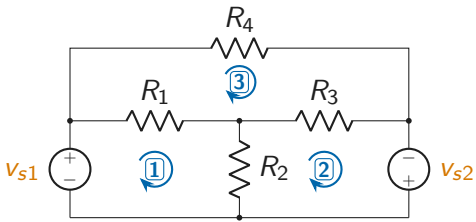
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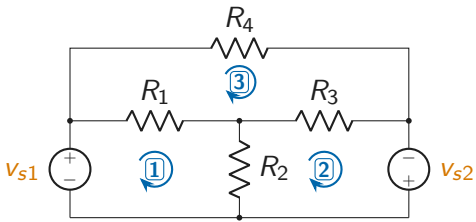
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A source met from $-$ to $+$ enters with a minus sign.

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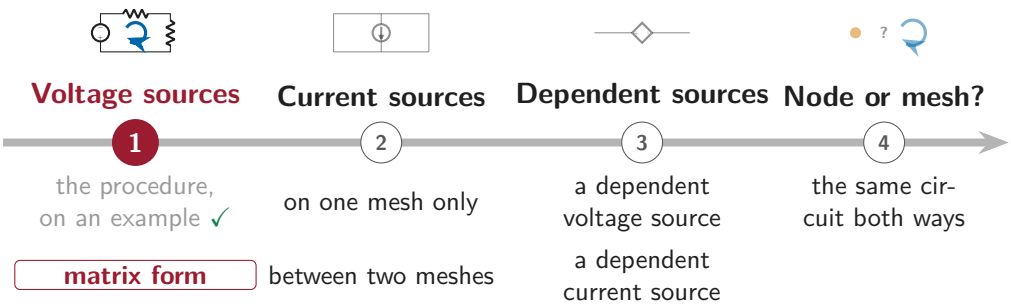
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Three equations, three unknowns.

Then Steps 2 and 3 give every branch current and voltage.

Roadmap



Each station adds a new kind of source; the last one compares the two methods.

Example 12.1 in matrix form

Collect the terms of each mesh:

$$\begin{aligned}(R_1 + R_2)\, i_{M1} - R_2\, i_{M2} - R_1\, i_{M3} &= v_{s1} \\ -R_2\, i_{M1} + (R_2 + R_3)\, i_{M2} - R_3\, i_{M3} &= v_{s2} \\ -R_1\, i_{M1} - R_3\, i_{M2} + (R_1 + R_3 + R_4)\, i_{M3} &= 0\end{aligned}$$

Question. Write the three equations as one matrix equation.

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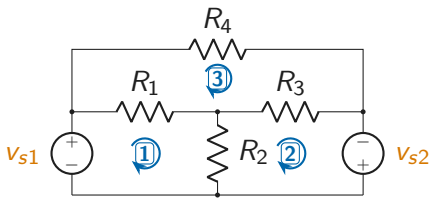
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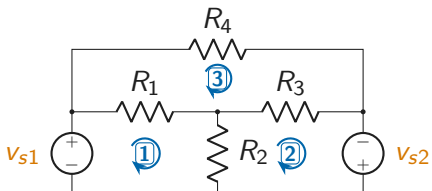
$\mathbf{v}_M$: the source voltages rising along each mesh.

Writing Z_M directly from the circuit

Question. Can we read Z_M from the circuit, without writing KVL?



Writing Z_M directly from the circuit

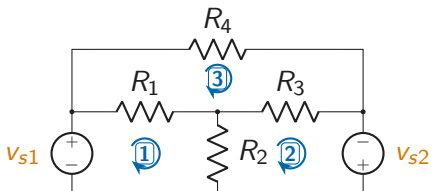


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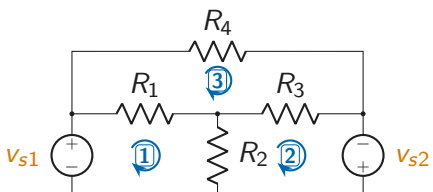
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Off-diagonal (k, j): minus the sum of the resistances shared by meshes k and j.

$(1, 2)$: $-R_2$.

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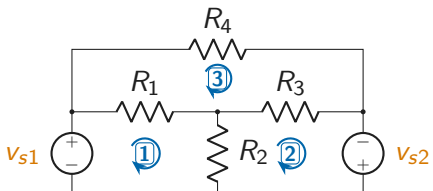
Off-diagonal (k, j): minus the sum of the resistances shared by meshes k and j.

$(1, 2)$: $-R_2$.

Right side k: the sum of the source voltages rising along mesh k.

row 1: v_{s1}, met from $-$ to $+$.

Writing Z_M directly from the circuit



Question. Can we read Z_M from the circuit, without writing KVL?

Diagonal (k, k): the sum of the resistances around mesh k.

$(3, 3)$: $R_1 + R_3 + R_4$.

Off-diagonal (k, j): minus the sum of the resistances shared by meshes k and j.

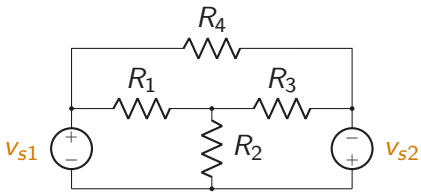
$(1, 2)$: $-R_2$.

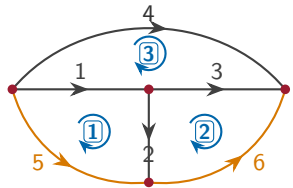
Right side k: the sum of the source voltages rising along mesh k.

row 1: v_{s1}, met from $-$ to $+$.

So Z_M is symmetric: entry (k, j) equals entry (j, k).

Z_M from the graph: the mesh matrix

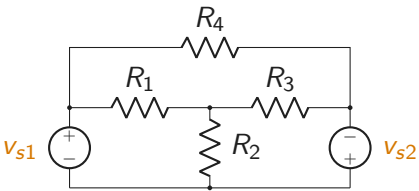


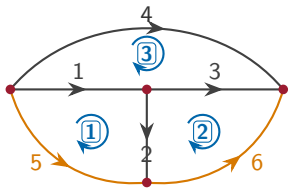


Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

Question. Write the mesh matrix M.

Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

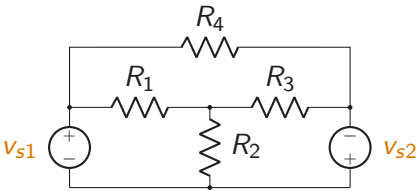
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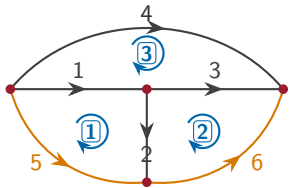
One row for each inner mesh.

Entry $+1$: the branch goes along the mesh; -1: against it.

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} & & & & & \\ & & & & & \\ & & & & & \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

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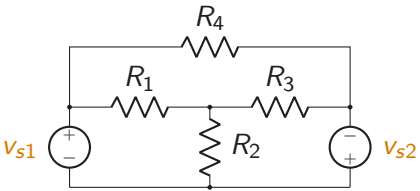
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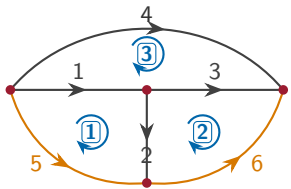
Entry $+1$: the branch goes along the mesh; -1: against it.

Row k: which branches are on mesh k?

$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & -1 & 0 \\ & & & & & \\ & & & & & \end{bmatrix} \end{array}$$

Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

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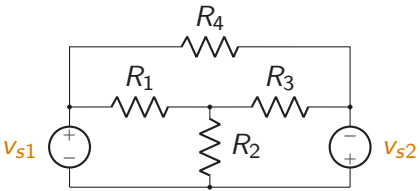
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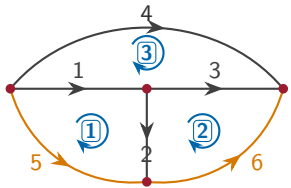
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Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

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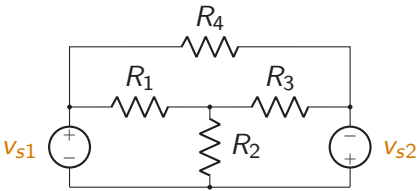
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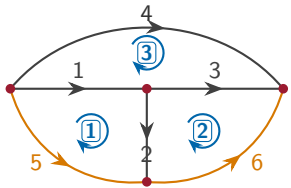
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Z_M from the graph: the mesh matrix





Branches 1–4: $R_1, \ldots, R_4$; 5, 6: v_{s1}, v_{s2}; a source branch points from $+$ to $-$.

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M_R: resistors M_S: sources

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Number the b_R resistors first, then the b_S voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

Z_M from the graph: KVL, KCL and Ohm's law in blocks

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KVL: $M\mathbf{v} = [\, M_R \;\; M_S \,] \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = M_R\mathbf{v}_R + M_S\mathbf{v}_S = \mathbf{0}$

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KCL: M^T splits into rows:

$$\mathbf{i} = M^T\mathbf{i}_M : \qquad \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = \begin{bmatrix} M_R^T \\ M_S^T \end{bmatrix} \mathbf{i}_M$$

b_R resistor rows: $\mathbf{i}_R = M_R^T\mathbf{i}_M$

b_S source rows: $\mathbf{i}_S = M_S^T\mathbf{i}_M$

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b_S source rows: $\mathbf{i}_S = M_S^T\mathbf{i}_M$

Only the resistor rows are needed: the source currents follow afterwards.

Z_M from the graph: KVL, KCL and Ohm's law in blocks

Number the b_R resistors first, then the b_S voltage sources:

$$M = [\, M_R \;\; M_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

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KCL: M^T splits into rows:

$$\mathbf{i} = M^T\mathbf{i}_M : \qquad \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = \begin{bmatrix} M_R^T \\ M_S^T \end{bmatrix} \mathbf{i}_M$$

b_R resistor rows: $\mathbf{i}_R = M_R^T\mathbf{i}_M$

b_S source rows: $\mathbf{i}_S = M_S^T\mathbf{i}_M$

Only the resistor rows are needed: the source currents follow afterwards.

Resistors: $\mathbf{v}_R = R\,\mathbf{i}_R$, $R = \mathrm{diag}(R_1, \ldots, R_{b_R})$, one resistance per resistor.

Z_M from the graph: the general result

Put Ohm's law and the resistor rows of KCL into KVL:

$$M_R \mathbf{v}_R + M_S \mathbf{v}_S = M_R\, R\, \underbrace{M_R^T \mathbf{i}_M}_{\mathbf{i}_R} + M_S \mathbf{v}_S = \mathbf{0} \quad \Rightarrow \quad M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

Z_M from the graph: the general result

Put Ohm's law and the resistor rows of KCL into KVL:

$$M_R \mathbf{v}_R + M_S \mathbf{v}_S = M_R\, R\, \underbrace{M_R^T \mathbf{i}_M}_{\mathbf{i}_R} + M_S \mathbf{v}_S = \mathbf{0} \quad \Rightarrow \quad M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

$$Z_M = M_R\, R\, M_R^T, \qquad \mathbf{v}_M = -M_S \mathbf{v}_S$$

Z_M from the graph: the general result

Put Ohm's law and the resistor rows of KCL into KVL:

$$M_R \mathbf{v}_R + M_S \mathbf{v}_S = M_R\, R\, \underbrace{M_R^T \mathbf{i}_M}_{\mathbf{i}_R} + M_S \mathbf{v}_S = \mathbf{0} \quad \Rightarrow \quad M_R\, R\, M_R^T\, \mathbf{i}_M = -M_S \mathbf{v}_S$$

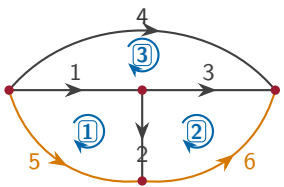
$$Z_M = M_R\, R\, M_R^T, \qquad \mathbf{v}_M = -M_S \mathbf{v}_S$$

$$Z_M^T = M_R\, R^T M_R^T = M_R\, R\, M_R^T = Z_M, \text{ since } R \text{ is diagonal.}$$

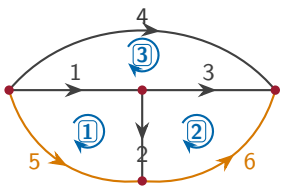
So Z_M is always symmetric for resistors and voltage sources.

Z_M from the graph: Example 12.1, the resistor currents

M_R^T: the resistor columns 1–4 of M, written as rows.



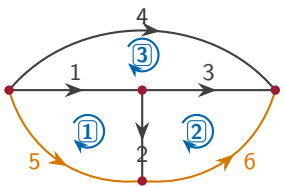
Z_M from the graph: Example 12.1, the resistor currents



M_R^T: the resistor columns 1–4 of M, written as rows.

$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix}$$

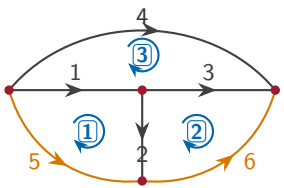
Z_M from the graph: Example 12.1, the resistor currents



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$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} \\ i_{M1} - i_{M2} \\ i_{M2} - i_{M3} \\ i_{M3} \end{bmatrix}$$

Z_M from the graph: Example 12.1, the resistor currents

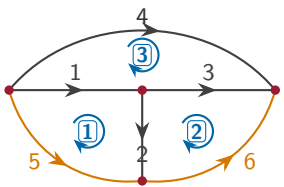


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Each resistor current: the mesh currents along its arrow minus those against it.

Z_M from the graph: Example 12.1, the resistor currents



M_R^T: the resistor columns 1–4 of M, written as rows.

$$\underbrace{\begin{bmatrix} i_{R_1} \\ i_{R_2} \\ i_{R_3} \\ i_{R_4} \end{bmatrix}}_{\mathbf{i}_R} = \underbrace{\begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{M_R^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} \\ i_{M1} - i_{M2} \\ i_{M2} - i_{M3} \\ i_{M3} \end{bmatrix}$$

Each resistor current: the mesh currents along its arrow minus those against it.

Then Ohm's law: $\mathbf{v}_R = R\,\mathbf{i}_R$, $v_{R_k} = R_k i_{R_k}$.

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

$$M_R\, R\, M_R^T = \begin{bmatrix} R_1 + R_2 & -R_2 & -R_1 \\ -R_2 & R_2 + R_3 & -R_3 \\ -R_1 & -R_3 & R_1 + R_3 + R_4 \end{bmatrix}$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

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Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

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$$-M_S \mathbf{v}_S = - \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}$$

Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

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Z_M from the graph: Example 12.1, the matrix

$$M_R\, R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}$$

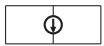
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The three rules read this product off, entry by entry.

Roadmap









Voltage sources ✓

the procedure, on an example ✓

matrix form ✓

Current sources 2

on one mesh only

between two meshes

Dependent sources 3

a dependent voltage source

a dependent current source

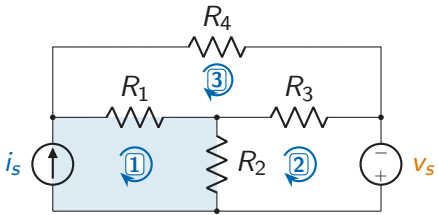
Node or mesh? 4

the same circuit both ways

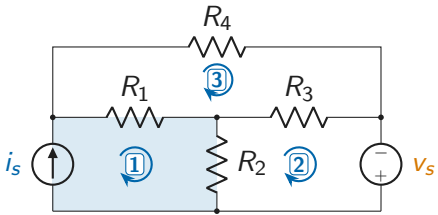
Each station adds a new kind of source; the last one compares the two methods.

Example 12.2: a current source on one mesh only

i_s is on mesh 1 only: $i_{M1} = i_s$ is known.



Example 12.2: a current source on one mesh only

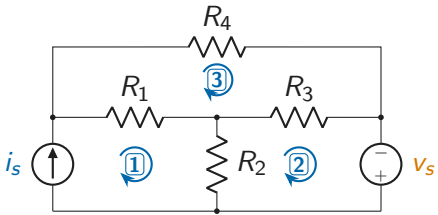


i_s is on mesh 1 only: $i_{M1} = i_s$ is known.

We do not write KVL around mesh 1: the voltage of i_s is not known.

The unknowns are i_{M2}, i_{M3}.

Example 12.2: a current source on one mesh only



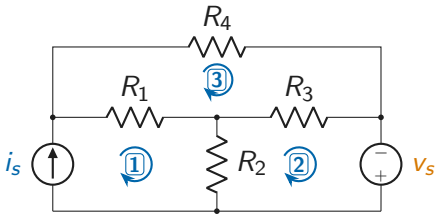
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Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

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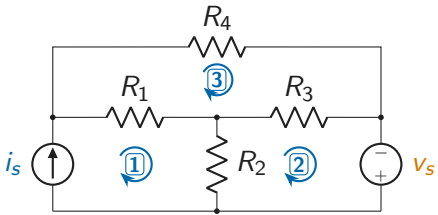
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Mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

Mesh 3:

$R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

Example 12.2: a current source on one mesh only



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We do not write KVL around mesh 1: the voltage of i_s is not known.

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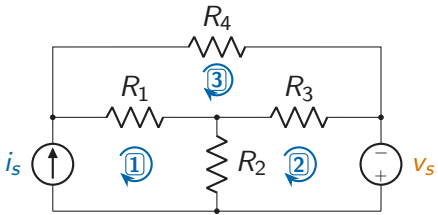
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$$\begin{bmatrix} R_2 + R_3 & -R_3 \\ -R_3 & R_1 + R_3 + R_4 \end{bmatrix} \begin{bmatrix} i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} v_s + R_2 i_s \\ R_1 i_s \end{bmatrix}$$

Example 12.2: a current source on one mesh only



i_s is on mesh 1 only: $i_{M1} = i_s$ is known.

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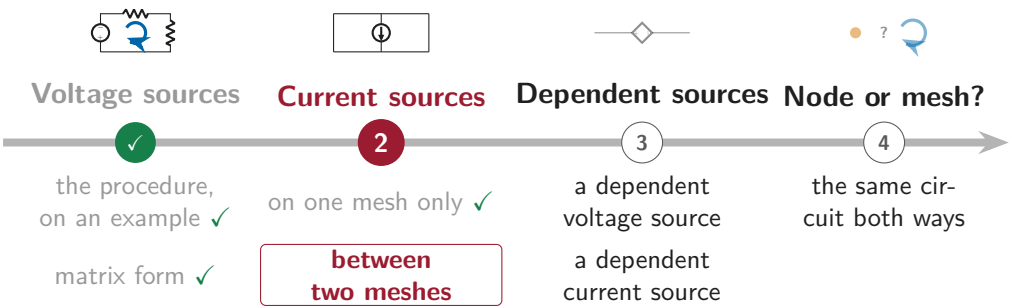
Mesh 3:

$R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

$$\begin{bmatrix} R_2 + R_3 & -R_3 \\ -R_3 & R_1 + R_3 + R_4 \end{bmatrix} \begin{bmatrix} i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} v_s + R_2 i_s \\ R_1 i_s \end{bmatrix}$$

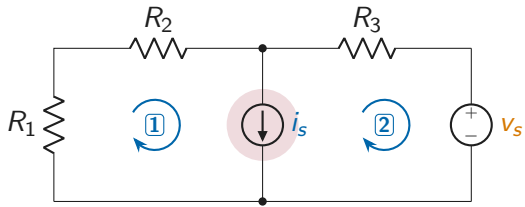
The known i_s moves to the right side.

Roadmap



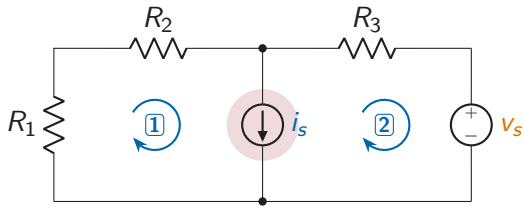
Each station adds a new kind of source; the last one compares the two methods.

Example 12.3: a current source between two meshes



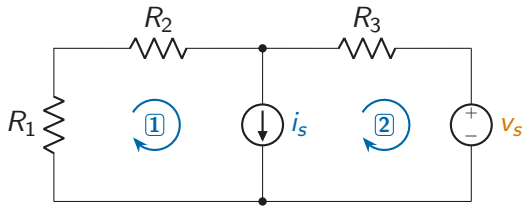
- **Question.** Write KVL around mesh 1: what is the voltage across i_s?

Example 12.3: a current source between two meshes



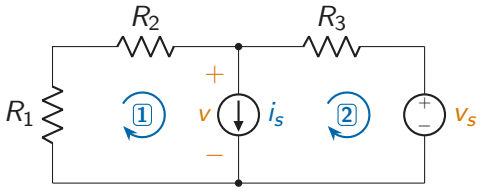
- **Question.** Write KVL around mesh 1: what is the voltage across i_s?
- It cannot be written with i_{M1} and i_{M2}: the law of a current source fixes its current, not its voltage.

Example 12.3: a current source between two meshes



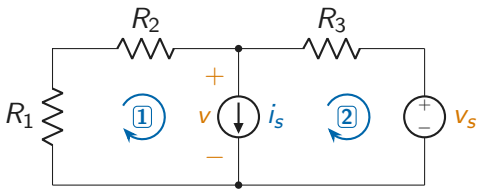
- **Question.** Write KVL around mesh 1: what is the voltage across i_s?
- It cannot be written with i_{M1} and i_{M2}: the law of a current source fixes its current, not its voltage.
- **Modified mesh current analysis** offers two ways out: Method I (its voltage is a new unknown) and Method II (a supermesh).

Method I: the voltage as an unknown



The voltage v of i_s is one more unknown.

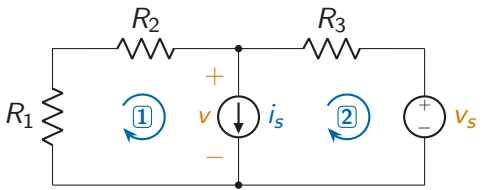
Method I: the voltage as an unknown



Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

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Method I: the voltage as an unknown

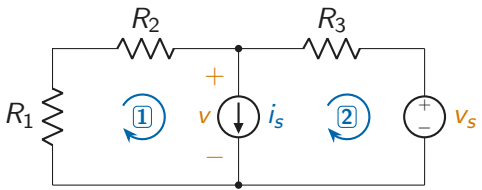


Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

Mesh 2: $R_3\, i_{M2} + v_s - v = 0$

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Method I: the voltage as an unknown



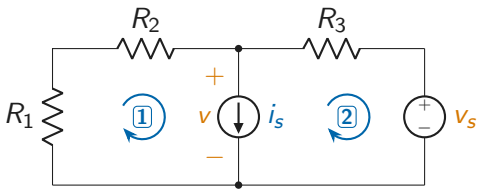
Mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

Mesh 2: $R_3\, i_{M2} + v_s - v = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

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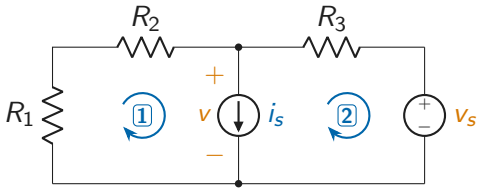
Mesh 2: $R_3\, i_{M2} + v_s - v = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

The voltage v of i_s is one more unknown.

The constraint: i_s carries i_{M1} down and i_{M2} up.

Method I: the voltage as an unknown



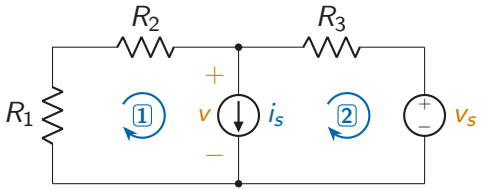
$$\begin{bmatrix} R_1+R_2 & 0 & 1 \\ 0 & R_3 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ -v_s \\ i_s \end{bmatrix}$$

The voltage v of i_s is one more unknown.

The constraint: i_s carries i_{M1} down and i_{M2} up.

Three unknowns, three equations; the last row is the constraint.

Method I: the voltage as an unknown



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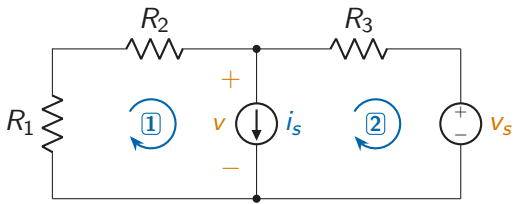
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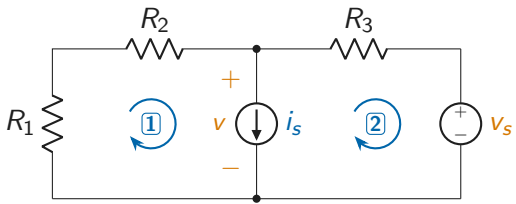
It works, but the supermesh avoids the extra unknown.

Method II: the supermesh



Question. Can we get rid of the unknown v?

Method II: the supermesh

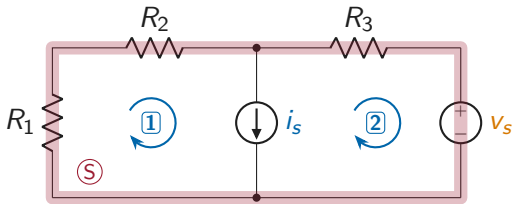


Question. Can we get rid of the unknown v?

Add the KVL equations of meshes 1 and 2.

v enters mesh 1 with $+$ and mesh 2 with $-$, so it cancels.

Method II: the supermesh



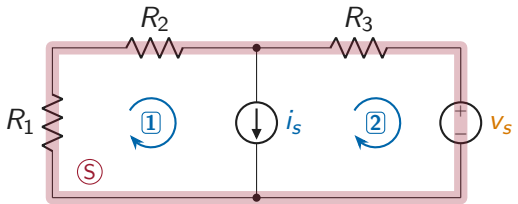
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This is KVL around the outer loop Ⓢ of the two meshes; it avoids i_s.

Method II: the supermesh



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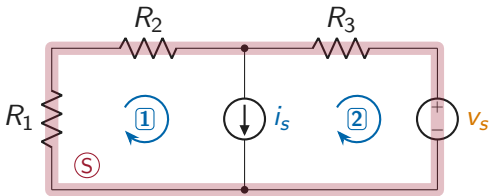
v enters mesh 1 with $+$ and mesh 2 with $-$, so it cancels.

This is KVL around the outer loop Ⓢ of the two meshes; it avoids i_s.

Supermesh

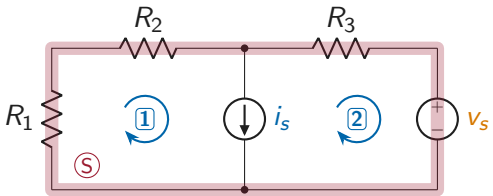
Write KVL around two meshes as if they were a single mesh, skipping the current source between them.

Method II: the equations



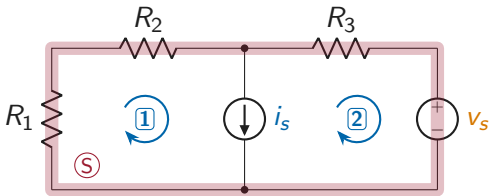
Question. Write KVL around the supermesh.

Method II: the equations



Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

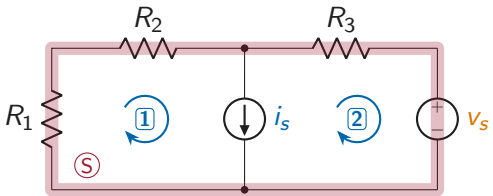
Method II: the equations



Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

Method II: the equations

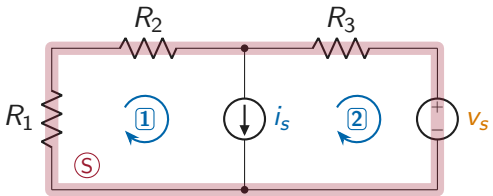


Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

Method II: the equations



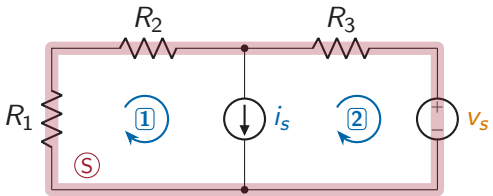
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Two unknowns: one fewer than Method I.

Method II: the equations



Supermesh: $(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0$

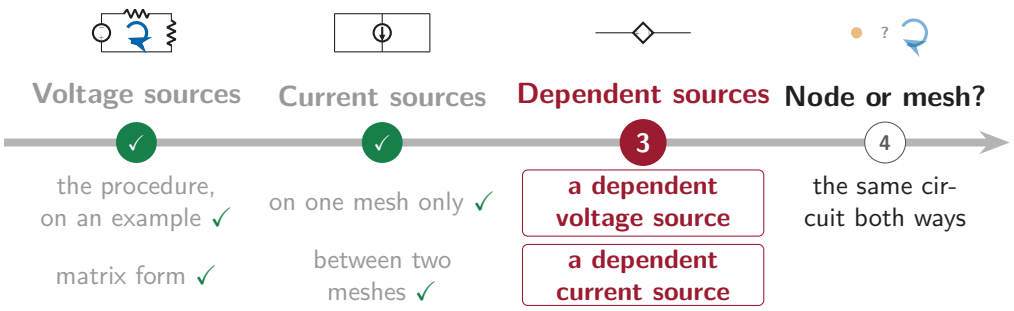
Constraint: $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}$$

Two unknowns: one fewer than Method I.

The matrix is not symmetric any more.

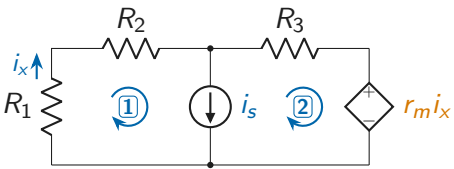
Roadmap



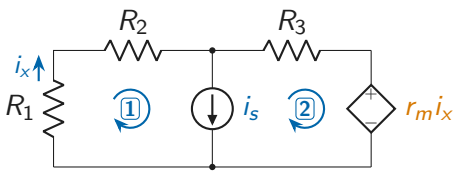
Each station adds a new kind of source; the last one compares the two methods.

Example 12.4: a dependent voltage source

Question. Write the mesh equations.



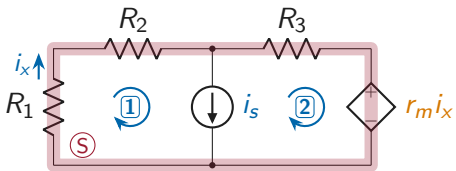
Example 12.4: a dependent voltage source



Question. Write the mesh equations.

Treat the dependent source as a voltage source of value $r_m i_x$.

Example 12.4: a dependent voltage source

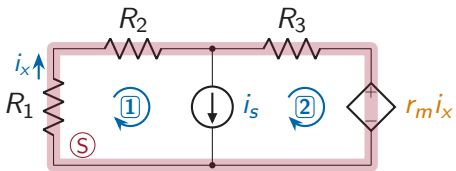


Question. Write the mesh equations.

Treat the dependent source as a voltage source of value $r_m i_x$.

i_s is between meshes 1 and 2: use the supermesh (S).

Example 12.4: a dependent voltage source



Question. Write the mesh equations.

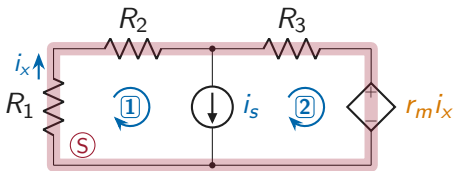
Treat the dependent source as a voltage source of value $r_m i_x$.

i_s is between meshes 1 and 2: use the supermesh Ⓢ.

Supermesh:

$$(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + r_m i_x = 0$$

Example 12.4: a dependent voltage source



Question. Write the mesh equations.

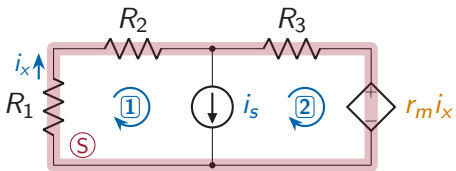
Treat the dependent source as a voltage source of value $r_m i_x$.

i_s is between meshes 1 and 2: use the supermesh Ⓢ.

Supermesh:
$(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + r_m i_x = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

Example 12.4: a dependent voltage source



Question. Write the mesh equations.

Treat the dependent source as a voltage source of value $r_m i_x$.

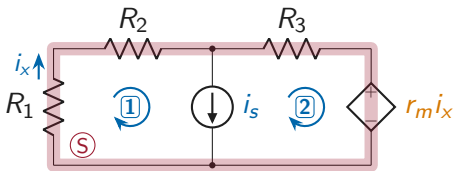
i_s is between meshes 1 and 2: use the supermesh Ⓢ.

Supermesh:
$(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + r_m i_x = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

Question. What is i_x in mesh currents?

Example 12.4: a dependent voltage source



Question. Write the mesh equations.

Treat the dependent source as a voltage source of value $r_m i_x$.

i_s is between meshes 1 and 2: use the supermesh Ⓢ.

Supermesh:
$(R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + r_m i_x = 0$

Constraint: $i_{M1} - i_{M2} = i_s$

Question. What is i_x in mesh currents?

The control: $i_x = i_{M1}$.

Example 12.4: matrix form

As if $r_m i_x$ were an independent source:

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -r_m i_x \\ i_s \end{bmatrix}$$

Example 12.4: matrix form

As if $r_m i_x$ were an independent source:

$$\begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -r_m i_x \\ i_s \end{bmatrix}$$

Question. Put $i_x = i_{M1}$: which entry changes?

Example 12.4: matrix form

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Put $i_x = i_{M1}$ and move $r_m i_{M1}$ to the left side:

$$\begin{bmatrix} R_1 + R_2 + r_m & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} 0 \\ i_s \end{bmatrix}$$

Example 12.4: matrix form

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$$i_{M1} = \frac{R_3\, i_s}{R_1 + R_2 + R_3 + r_m}, \qquad i_{M2} = i_{M1} - i_s$$

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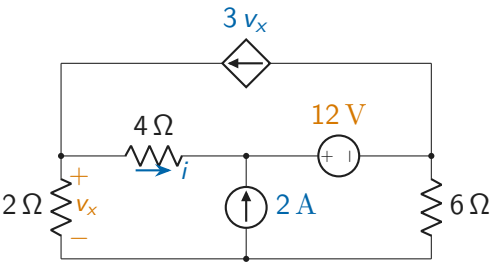
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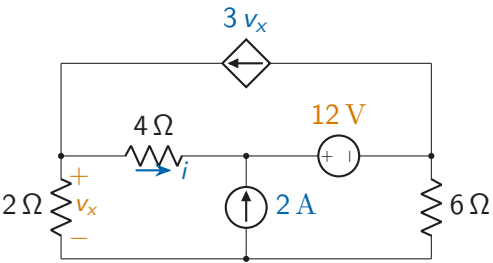
The dependent source changes one entry of the matrix.

Example 12.5: a dependent current source

Question. Find i, the current of the $4\,\Omega$.



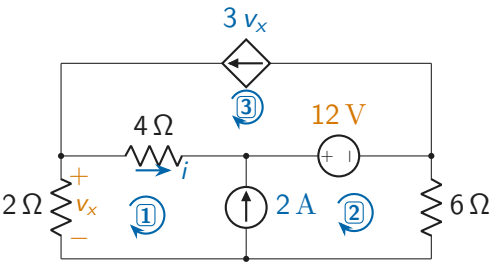
Example 12.5: a dependent current source



Question. Find i, the current of the 4 Ω.

Treat the dependent source as a current source of value $3\,v_x$.

Example 12.5: a dependent current source

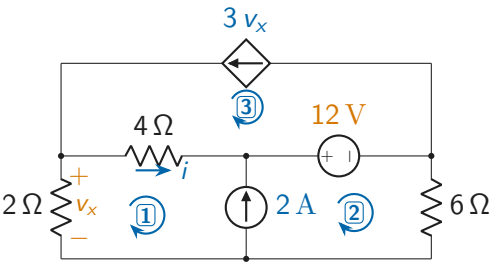


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Three inner meshes: unknowns i_{M1}, i_{M2}, i_{M3}.

Example 12.5: a dependent current source



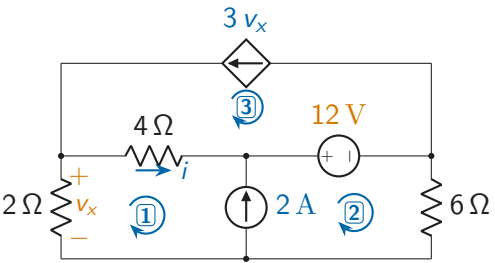
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Example 12.5: a dependent current source



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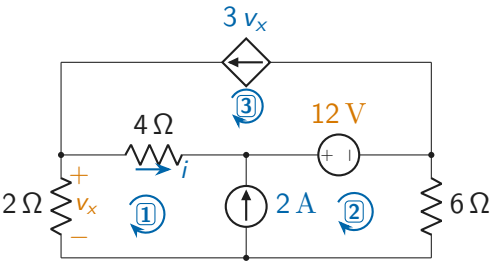
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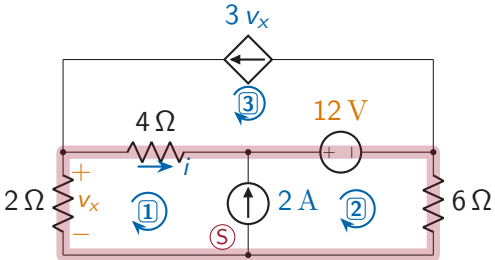
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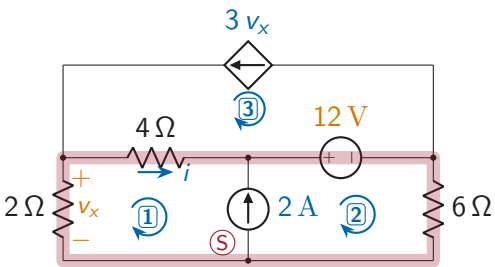
Equations: KVL around the supermesh and two constraints.

Example 12.5: the equations



Question. Write KVL around the supermesh.

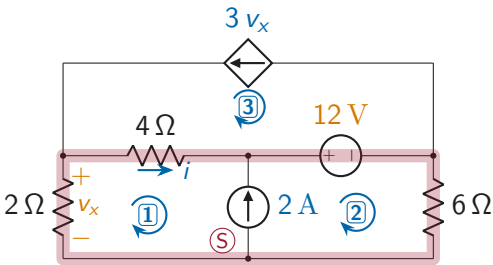
Example 12.5: the equations



Supermesh:

$$2\,i_{M1} + 4(i_{M1} - i_{M3}) + 12 + 6\,i_{M2} = 0 \quad (1)$$

Example 12.5: the equations

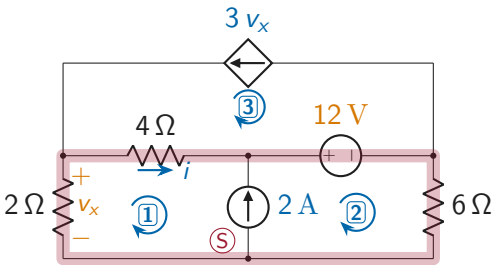


Supermesh:

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The $2\,\mathrm{A}$ source: $i_{M2} - i_{M1} = 2 \quad (2)$

Example 12.5: the equations



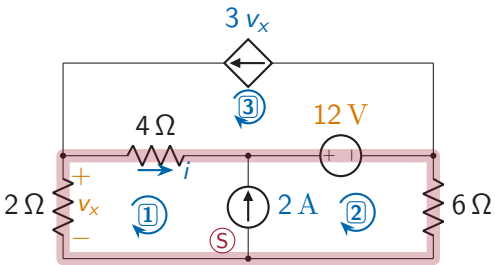
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Example 12.5: the equations



Supermesh:

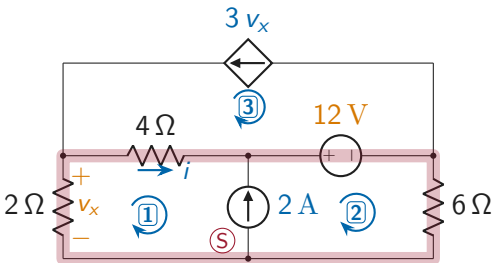
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Question. What is v_x in mesh currents?

Example 12.5: the equations



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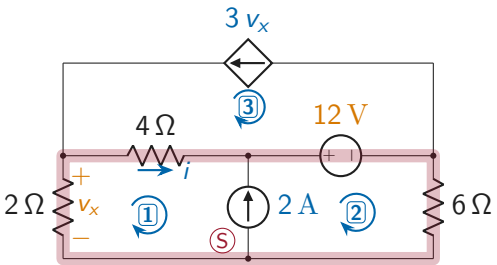
The 2 A source: $i_{M2} - i_{M1} = 2$ (2)

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i_{M1} flows up through the 2 Ω, from − to +: $v_x = -2\,i_{M1}$.

Example 12.5: the equations



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Question. What is v_x in mesh currents?

i_{M1} flows up through the 2 Ω, from − to +: $v_x = -2\,i_{M1}$.

Constraint: $i_{M3} = 6\,i_{M1} \quad (3)$

Example 12.5: matrix form

Collect the terms of (1), (2) and (3):

$$\begin{bmatrix} 6 & 6 & -4 \\ -1 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} -12 \\ 2 \\ 0 \end{bmatrix}$$

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Rows: KVL around the supermesh, the $2\,\mathrm{A}$ source, the dependent source.

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Rows: KVL around the supermesh, the $2\,\mathrm{A}$ source, the dependent source.

Solve: $i_{M1} = 2\,\mathrm{A}$, $i_{M2} = 4\,\mathrm{A}$, $i_{M3} = 12\,\mathrm{A}$.

Example 12.5: matrix form

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Rows: KVL around the supermesh, the $2\,\mathrm{A}$ source, the dependent source.

Solve: $i_{M1} = 2\,\mathrm{A}$, $i_{M2} = 4\,\mathrm{A}$, $i_{M3} = 12\,\mathrm{A}$.

$$i = i_{M1} - i_{M3} = 2 - 12 = -10\,\mathrm{A}$$

Dependent sources

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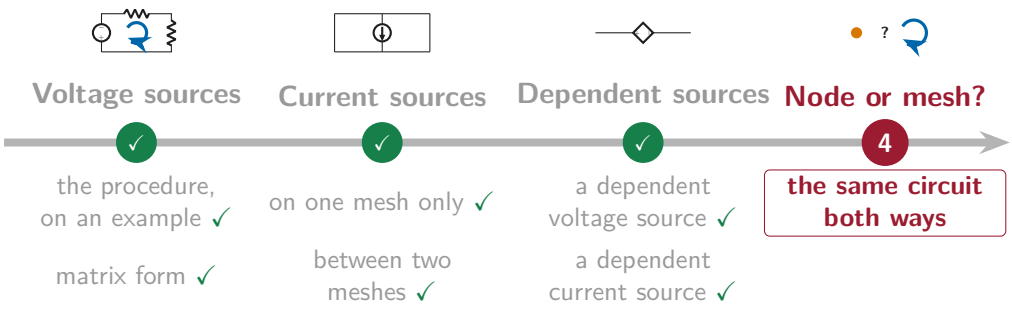
- A dependent source is handled like an independent one of the same kind.
- Its control variable (i_x, v_x) is written with mesh currents: this gives the constraint equation.

Remark

If a dependent current source is on mesh k only, do not write KVL around mesh k: its voltage is not known.

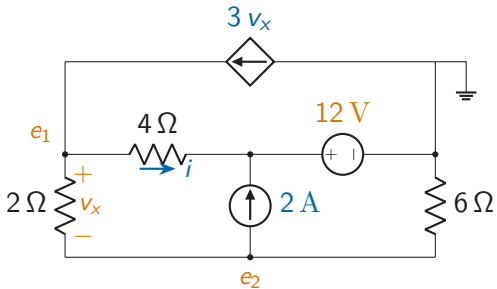
Write the constraint that the dependent source gives instead.

Roadmap



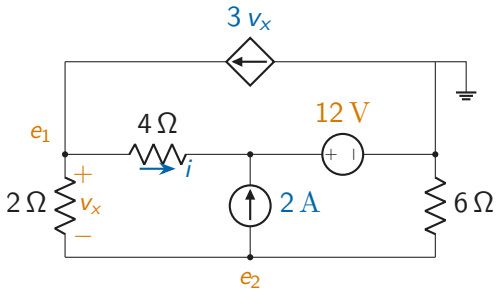
Each station adds a new kind of source; the last one compares the two methods.

Example 12.5 with node voltages



Question. Solve the same circuit with node voltages.

Example 12.5 with node voltages



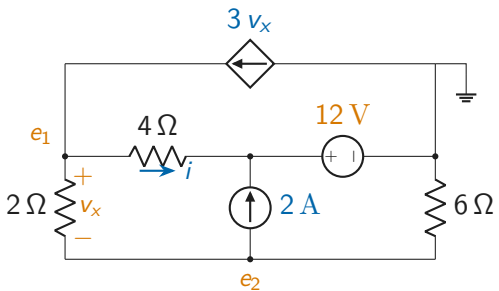
Question. Solve the same circuit with node voltages.

Take the right node as the ground.

The 12 V source goes from the middle node to the ground: that node is at 12 V.

Unknowns: e_1, e_2.

Example 12.5 with node voltages



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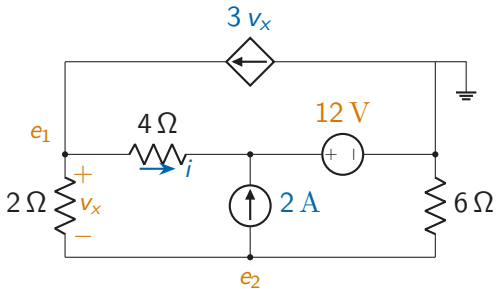
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Node 1: $\dfrac{e_1 - e_2}{2} + \dfrac{e_1 - 12}{4} - 3\,v_x = 0$

Example 12.5 with node voltages



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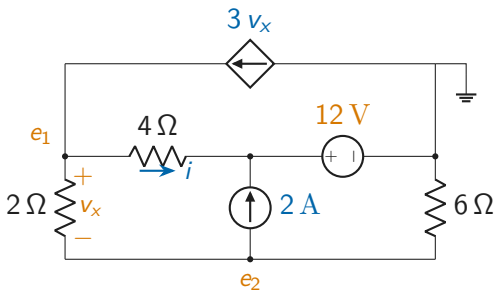
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Constraint: $v_x = e_1 - e_2$

Example 12.5 with node voltages: matrix form

Put $v_x = e_1 - e_2$ and collect the terms:

$$\begin{bmatrix} -\frac{9}{4} & \frac{5}{2} \\ -\frac{1}{2} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

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$$i = \frac{e_1 - 12}{4} = \frac{-28 - 12}{4} = -10\,\mathrm{A}$$

The same answer, as it must be.

Node or mesh?

Question. For Example 12.5, which method needs fewer equations?

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Mesh currents

$\ell = 3$ unknowns.

Each current source fixes one of them:
$3 - 2 = 1$ KVL equation is left.

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Mesh currents

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Each current source fixes one of them:
$3 - 2 = 1$ KVL equation is left.

Node voltages

$n - 1 = 3$ unknowns.

Each voltage source fixes one of them:
$3 - 1 = 2$ KCL equations are left.

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Mesh currents

$\ell = 3$ unknowns.

Each current source fixes one of them:
$3 - 2 = 1$ KVL equation is left.

Node voltages

$n - 1 = 3$ unknowns.

Each voltage source fixes one of them:
$3 - 1 = 2$ KCL equations are left.

Tip. Count before you start: mesh, ℓ minus the current sources; node, $n - 1$ minus the voltage sources.

Mesh current analysis needs a planar circuit; node voltage analysis always works.

Summary

In the circuit	What to do
resistors and voltage sources	KVL around every inner mesh: $Z_M \mathbf{i}_M = \mathbf{v}_M$, from three simple rules; $Z_M = M_R R M_R^T$ is symmetric
a current source on one mesh only	that mesh current is known; no KVL around that mesh
a current source between two meshes	Method I: its voltage as an unknown, plus the constraint; Method II: KVL around the supermesh, plus the constraint
a dependent source	write its control variable with mesh currents: a constraint equation
node or mesh?	count the equations; mesh needs a planar circuit

After the mesh currents are found: $\mathbf{i} = M^T \mathbf{i}_M$, then the branch voltages from the element laws.